

Krishna Sandeep Kothapalli

DevOps Cloud Engineer

Location: Dallas, TX

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Professional Summary:

- Distinguished DevOps engineer with 13+ years of experience in CI/CD, Cloud Infrastructure management, and automation, with a strong background in **Python and Java**.
- Expertise in multi-cloud infrastructure design and management across **AWS, Azure, and GCP**, optimizing cost, security, and high availability for large-scale deployments.
- Designed and managed **AWS** infrastructure with services like **EC2, S3, RDS, EKS, Lambda**, ensuring **fault tolerance, high availability, and cost optimization**.
- Hands-on experience in Azure development, working with **Azure App Services, Azure SQL Database, Azure AD, AKS, and Azure Storage**, deploying multi-tier solutions using **Azure Resource Manager** and Virtual Machines for high availability, fault tolerance, and auto-scaling.
- Designed and deployed secure, scalable infrastructure in **GCP**, leveraging **VPC networking, IAM security, Google Kubernetes Engine (GKE), Cloud Load Balancing, Cloud Storage, and hybrid cloud connectivity** using **VPN and Interconnect**.
- Implemented **Kubernetes** security best practices across **AWS EKS, Azure AKS, and GCP GKE**, including **RBAC, network policies, and container image scanning tools** such as **Trivy and Aqua Security**.
- Managed **IAM** roles, users, and groups across **AWS, Azure, and GCP**, creating **custom policies** to enhance **security and access management**.
- Automated infrastructure provisioning deployment using **Jenkins, Terraform, and Ansible**, improving operational efficiency and reducing manual effort across multi-cloud environments.
- Designed and managed CI/CD pipelines in **Jenkins, GitHub Actions**, integrating **Maven, AWS CLI, Nexus, Docker, ArgoCD, Helm and Kubernetes** to streamline delivery processes.
- Developed serverless REST web services using **AWS Lambda, API Gateway, and Python**, alongside **Azure Functions** and **Google Cloud Functions**, deploying workloads across multiple cloud platforms.
- Implemented monitoring and observability solutions using **Prometheus, Grafana, ELK, Datadog, Azure Monitor, CloudWatch, and Google Cloud Operations**, integrating real-time metrics, alerts, and dashboards for proactive monitoring.
- Experienced in multi-cloud architecture and automation, analyzing requirements and designing comprehensive solutions for deployment and operation across **AWS, Azure, and GCP**.

Certifications: -

- [Microsoft Certified: Azure DevOps Engineer Expert](#)
- [Google Cloud Certified - Associate Cloud Engineer](#)
- [Microsoft Certified: Azure Developer Associate](#)
- [HashiCorp Certified: Terraform Associate](#)
- [Microsoft Certified: Azure Fundamentals](#)
- [AWS Certified: Cloud Practitioner](#)

Technical Skills:

Technologies:

- **Cloud Platforms:** Azure, AWS, GCP
- **Containerization:** Docker, Kubernetes, EKS, GKE, AKS

- **CI/CD Tools:** Jenkins, ArgoCD, Ansible, Nexus, Artifactory, SonarQube, Maven, ANT, Gradle, Terraform
- **Version Control Systems:** GitHub, Bitbucket, SVN
- **Development Languages:** Python, Java, PowerShell
- **Monitoring and Logging:** Datadog, Prometheus, Grafana, Cloud Watch, Nagios, Azure Monitor, Azure Log
- **Project Management Tools:** JIRA, ServiceNow
- **Databases:** Oracle, PostgreSQL, MySQL, MongoDB

Professional Experience:

Client: FedEx, Dallas, TX

Apr 2016 – Present

DevOps Cloud Engineer

- Designed highly available, fault-tolerant solutions across **AWS, Azure** and **GCP** using Terraform, achieving **30% cost savings** through optimized resource allocation and usage.
- Implemented hybrid cloud connectivity between **AWS, Azure**, and **GCP** using **Direct Connect**, **ExpressRoute**, and Cloud **Interconnect**, ensuring secure, low-latency communication between on-prem and cloud workloads while optimizing network costs and performance.
- Developed and maintained reusable, parameterized **Terraform** modules for **AWS, Azure**, and **GCP** automating infrastructure deployment while ensuring consistency, reusability, and reducing configuration drift.
- Planned and executed secure and scalable networking solutions in **Google Cloud Platform (GCP)**, including **VPC peering**, firewall rules, Cloud Load Balancing, and private connectivity using Cloud **Interconnect** and **VPN** for hybrid cloud architectures.
- Architected and managed production-grade Kubernetes (**GKE, EKS** and **AKS**) clusters, leveraging **Helm charts** and **ArgoCD** for automated versioned deployments, rollbacks, and GitOps workflows **reducing deployment times by 40%**.
- Enforced **Kubernetes** security best practices by implementing **RBAC**, network policies, pod security policies, and image scanning with **Trivy** and **Aqua Security**, proactively reducing vulnerabilities and enhancing container workload protection.
- Deployed **Kubernetes** Operators to manage custom resources, automate infrastructure tasks, and enhance self-healing capabilities, improving resilience and scalability of **containerized workloads**.
- Strengthened cloud security by implementing encryption at rest and in transit across AWS, Azure, and GCP using AWS **KMS**, **Secrets Manager**, **Azure Key Vault**, TLS, and **Azure Disk Encryption** to safeguard sensitive data.
- Architected **cross-region disaster recovery** with a Recovery Time Objective (RTO) under 10 minutes using **S3** replication, **DynamoDB** global tables, and Azure Geo-Redundant Storage.
- Enhanced data security by implementing **encryption at rest and in transit** using AWS KMS, Secrets Manager, Azure Key Vault, TLS, and Azure Disk Encryption.
- Improved distributed tracing and fault diagnosis across microservices using **AWS X-Ray** and Azure Application Insights, and created actionable dashboards in **Grafana** for performance visibility.
- Integrated and optimized multi-cloud database solutions across **PostgreSQL, MongoDB, Oracle**, and **MySQL**, implementing automated backup strategies, replication, and performance tuning to enhance scalability, high availability, and disaster recovery.
- Enabled compliance tracking with automated alerts using **AWS Config**, **Azure Policy**, and Sentinel to maintain regulatory adherence (SOC 2, HIPAA).

- Engineered end-to-end DevSecOps pipelines integrating **Jenkins**, **ArgoCD**, **GitHub Actions**, **SonarQube**, and **OWASP Dependency-Check**, automating static code analysis, vulnerability scanning, and compliance validation to enforce security at every stage of the **CI/CD** lifecycle.
- Developed a unified monitoring strategy using **Prometheus**, **Grafana**, **CloudWatch**, **Azure Monitor**, and **Datadog** implementing custom metrics, log aggregation, and automated alerting to proactively identify and resolve performance issues.

Client: Cigna, Chattanooga, TN

Sep 2014 – Mar 2016

DevOps Engineer

- Developed and streamlined automated build and deployment processes, enhancing user experience and establishing robust continuous integration
- Led the implementation of CI/CD processes using Jenkins, automating routine tasks with Python and Shell scripts
- Configured and administered **Jenkins**, and implemented build frameworks with **Maven** for efficient project builds and artifact management.
- Developed an automation framework for deploying applications to cloud environments, ensuring seamless deployment.
- Implemented **AWS** solutions including **EC2**, **S3**, **RDS**, **EBS**, **Elastic Load Balancers**, and **Auto Scaling** groups for optimized performance and scalability.
- Managed branching, tagging, and release activities using **Git**, ensuring effective version control and collaboration.
- Automated and deployed environments using **Ansible** on **AWS**, integrated with **Jenkins** for testing.
- Managed **Docker** containers, including creating snapshots, attaching to running containers, removing images, and handling directory structures.

Client: State Farm, Bloomington, IL

Aug 2011 – Aug 2014

Java Developer

- Developed **Java** applications using **Spring IOC** and **MVC** frameworks to ensure modularity and ease of maintenance.
- Created **DAO** layers with **Spring JDBC Template** for seamless database interaction with **Oracle**.
- Designed and implemented **RESTful** web services with **JAX-RS (Restlet)** for efficient communication between systems.
- Automated build and deployment processes using **Jenkins**, **IBM UrbanCode Deploy**, **Apache ANT**, and **Maven** scripts.
- Conducted **JUnit** and **EasyMock** for comprehensive unit testing, ensuring code quality and reliability.
- Managed source code with **Tortoise SVN**, performing merges and branch management efficiently.

Education: M.S in Computer Science, Texas A&M University Kingsville, Texas - 2011